

ALEJANDRA MARULANDA

Telematics Engineer

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Senior telematics engineering student focused on telecommunications, IoT, and distributed systems. Experienced in wireless networks, antenna design, and IoT architectures applied to space-analog and environmental projects. Interested in advancing in communications, cybersecurity, and network infrastructure, combining theory and hands-on implementation to build efficient and reliable solutions.

EXPERIENCE

Database Management (Social service) - Agencia Espacial Mexicana Jan 2024 - jun 2024

- Collaborated in the conceptual and logical design of a database for inventory management.
- Contributed to technical documentation, including conference transcription and English-Spanish translation.

Application Implementation - WordStaff USA Apr 2023 - Oct 2023

- Customized and adapted Forms.app to meet the company's operational requirements.
- Collaborated with administrative and technical teams to improve data collection and process efficiency.
- Documented procedures and provided user support to ensure successful tool adoption.

RELEVANT PROJECTS

Antenna Design & Wireless Monitoring Project – UPIITA-IPN Jan 2025- Jul 2025

- Application of transmission line and antenna theory (gain, impedance matching, operating frequency, radiation patterns) to design and prototype 2.4 GHz WiFi antennas (biquad and cloverleaf).
- Fabrication and testing of antenna prototypes for a wireless water-quality monitoring system deployed in the Xochimilco lake area.
- Performance characterization using a Vector Network Analyzer (VNA) and spectrum analyzer to evaluate return loss, bandwidth, and overall link performance.

Research Project - UPIITA-IPN Jan 2025 - Present

Research project “Design and Simulation of a Distributed IoT Network for Monitoring and Automatic Control in Analog Space Habitats”

- Development of a distributed IoT system using ESP32, Raspberry Pi, WiFi, MQTT, and ESP-NOW.
- Implementation of low-latency, fault-tolerant data links for sensor networks in constrained environments.
- Development of network topology, routing, and control strategies for reliable real-time monitoring in space-analog conditions.

Thesis Project – Universidad ICEL

Sep 2025- Present

Thesis "Basic Cybersecurity in Mexico City (2025): Preventing Common Attacks for Users with Limited Digital Literacy"

- Developing a technical guide in accessible language to help users with limited digital literacy understand and prevent common cyber attacks in Mexico.
- Focused on practical strategies against threats such as phishing, malware, SIM swapping, and social engineering.
- Aims to reduce user vulnerability and promote safer digital practices through clear, easy-to-follow guidance.

EDUCATION

UPIITA - IPN

2018 - Jan 2027

Bachelor's Degree in Telematics Engineering

Universidad ICEL

2020 - 2025

Bachelor's Degree in Computer Systems Engineering

ECCC

2015 - 2018

Technical degree in administrative computing

CERTIFICATIONS

- Introduction to Cloud Computing – IBM, 2023
- Introduction to Web Development with HTML, CSS, JavaScript – IBM, 2023
- Microsoft Office Suite proficiency certified – Testing Program 2018

HARD SKILLS

- Cellular systems (2G–5G, frequency reuse, network concepts)
- Antenna theory (gain, operating frequency, bandwidth, radiation patterns)
- Knowledge of computer networks, TCP/IP and OSI models
- Experience with Cisco Packet Tracer for network design and simulation
- Experience with Linux systems (Ubuntu, Kali)
- Proficient in Microsoft Office Suite
- Programming in C
- Web development with HTML, CSS, JavaScript, Node.js
- Skilled in SQL and MySQL
- Data structures management
- Familiar with MEAN (MongoDB, Express, Angular, Node.js) and MERN (MongoDB, Express, React, Node.js) stacks
- Scrum and Agile knowledge

SOFT SKILLS

- Strong communication skills
- Problem-solving
- Team collaboration
- Initiative
- Fast learner

LANGUAGES

- Spanish - Native
- English - Advanced
- Mandarin Chinese - Basic